



04 — Curvilinear Motion: Acceleration Without Changing Speed

Kleppner: Ch 1.8~1.10 | Time: 75 min

Core Question: Can acceleration exist even when speed stays exactly the same?

Scene: A Child on a Merry-Go-Round

A child rides a merry-go-round at a steady speed. The speed never changes — yet the child *feels* pushed outward, pressed against the rail.

If speed is constant, where does this "force" come from? The answer: **the direction of the velocity vector is changing**. And changing a vector — even just its direction — is acceleration.

✗ What If You Try With Algebra Alone?

The Paradox of Uniform Circular Motion

An object moves in a circle of radius r at constant speed v . Since $|\vec{v}| = v$ never changes, you might think $\vec{a} = 0$.

But that's wrong. Let's try to see why.

Attempt: Approximate the Direction Change

After a short time Δt , the object has moved an angle $\Delta\theta = v\Delta t/r$ along the circle. The velocity vectors at the start and end have the **same length** v , but their directions differ by $\Delta\theta$.

Draw the two velocity vectors tail-to-tail. The change $\Delta\vec{v}$ is the vector connecting their tips. For small $\Delta\theta$, the triangle is isosceles with two sides of length v and a small angle $\Delta\theta$ between them:

$$\frac{|\Delta \vec{v}|}{v} \approx \Delta \theta \quad (\text{similar triangles, but an approximation!})$$

So:

$$|\Delta \vec{v}| \approx v \Delta \theta = v \cdot \frac{v \Delta t}{r} = \frac{v^2}{r} \Delta t$$

$$|\vec{a}| \approx \frac{|\Delta \vec{v}|}{\Delta t} \approx \frac{v^2}{r}$$

This looks like it should be exact — but we used a geometric approximation ($\sin \theta \approx \theta$ for small angles). The exact relation is $|\Delta \vec{v}| = 2v \sin(\Delta \theta/2)$. Only in the limit $\Delta t \rightarrow 0$ does the approximation become equality:

$$|\vec{a}| = \lim_{\Delta t \rightarrow 0} \frac{2v \sin(v \Delta t/2r)}{\Delta t} = \lim_{\Delta t \rightarrow 0} \frac{2v \cdot (v \Delta t/2r)}{\Delta t} = \frac{v^2}{r}$$

Without the limit, v^2/r is merely an approximation.

Resolved Through Limits

The Exact Derivation

From the geometry of the circle: position $\vec{r}(t) = (r \cos \omega t, r \sin \omega t)$ with $\omega = v/r$. Take the change-in-a-blink twice:

$$\vec{v} = \frac{d\vec{r}}{dt} = (-r\omega \sin \omega t, r\omega \cos \omega t)$$

$$\vec{a} = \frac{d\vec{v}}{dt} = (-r\omega^2 \cos \omega t, -r\omega^2 \sin \omega t) = -\omega^2 \vec{r}$$

Magnitude: $|\vec{a}| = r\omega^2 = r(v/r)^2 = v^2/r$. Direction: always toward the center (the negative sign in $-\omega^2 \vec{r}$).

This is exact — no approximations needed. **The limit handled the direction change precisely.**

The General Decomposition: Tangential + Normal

For *any* curved path (not just circles), acceleration splits into two perpendicular parts:

$$\vec{a} = \frac{dv}{dt} \hat{t} + \frac{v^2}{\rho} \hat{n}$$

Component	Cause	Effect
$\frac{dv}{dt} \hat{t}$ (tangential)	Change in speed	Speeds up or slows down
$\frac{v^2}{\rho} \hat{n}$ (normal/ centripetal)	Change in direction	Curves the path, always toward the inside of the turn

ρ is the **radius of curvature** — at any point on a curve, you can fit a circle that matches the curve's bend. A sharp turn has small ρ (large centripetal acceleration). A straight line has $\rho = \infty$ (zero centripetal acceleration).

The Two Components Are Independent

A car can brake while turning: $\frac{dv}{dt} < 0$ (tangential, slowing down) while $\frac{v^2}{\rho} > 0$ (normal, holding the curve). The total acceleration is the vector sum.



Examples

Example 1: Uniform Circular Motion

$r = 2 \text{ m}$, $v = 4 \text{ m/s}$. Find the centripetal acceleration.

$$a_c = \frac{v^2}{r} = \frac{16}{2} = 8 \text{ m/s}^2$$

The direction is toward the center at every instant. In terms of $g = 10 \text{ m/s}^2$: $a_c = 0.8g$.

Example 2: Car on a Curve

A car takes a turn of radius 100 m at 20 m/s (72 km/h).

$$a_c = \frac{400}{100} = 4 \text{ m/s}^2 \approx 0.4g$$

The driver feels pushed sideways with about 40% of their weight. At 30 m/s: $a_c = \frac{900}{100} = 9 \text{ m/s}^2 \approx 0.9g$ — near the limit of tire grip.

Example 3: Radius of Curvature From a Trajectory

$y = x^2$ (a parabola). At $x = 1$, what is the radius of curvature? Formula: $\rho = \frac{(1 + (y')^2)^{3/2}}{|y''|}$.

$$y' = 2x, y'' = 2. \text{ At } x = 1: \rho = \frac{(1 + 4)^{3/2}}{2} = \frac{5\sqrt{5}}{2} \approx 5.59 \text{ m.}$$

At $x = 0$ (the vertex): $y' = 0$, so $\rho = \frac{1^{3/2}}{2} = \frac{1}{2} \text{ m}$ — the tightest bend is at the vertex.

Example 4: Speed-Up While Turning

$v(t) = 3t$ (linearly increasing speed), $\rho = 9 \text{ m}$ (constant radius).

Tangential: $a_t = \frac{dv}{dt} = 3 \text{ m/s}^2$. Normal: $a_n = \frac{v^2}{\rho} = \frac{9t^2}{9} = t^2$.

At $t = 2$: $a_t = 3$, $a_n = 4$, total $a = \sqrt{3^2 + 4^2} = 5 \text{ m/s}^2$.

Example 5: Tangential + Normal Decomposition

$\vec{r}(t) = (t^2, t^3)$. At $t = 1$, find the radius of curvature.

$\vec{v} = (2t, 3t^2)$, $\vec{a} = (2, 6t)$. At $t = 1$: $\vec{v} = (2, 3)$, $|\vec{v}| = \sqrt{13}$, $\vec{a} = (2, 6)$.

Tangential component: $a_t = \frac{\vec{v} \cdot \vec{a}}{|\vec{v}|} = \frac{4+18}{\sqrt{13}} = \frac{22}{\sqrt{13}}$.

Normal component: $a_n = \sqrt{|\vec{a}|^2 - a_t^2} = \sqrt{40 - \frac{484}{13}} = \sqrt{\frac{36}{13}} = \frac{6}{\sqrt{13}}$.

Then $\rho = \frac{v^2}{a_n} = \frac{13}{6/\sqrt{13}} = \frac{13\sqrt{13}}{6} \approx 7.8 \text{ m}$.

Example 6: Full Acceleration With Both Components

$\vec{r}(t) = (4 \cos t, 4 \sin t, 3t)$. This is a helix — circular in xy , rising in z .

$$\vec{v} = (-4 \sin t, 4 \cos t, 3), |\vec{v}| = \sqrt{16 + 9} = 5 \text{ (constant speed!).}$$

$$\vec{a} = (-4 \cos t, -4 \sin t, 0), |\vec{a}| = 4.$$

Since speed is constant, $a_t = 0$. All acceleration is centripetal: $a_n = 4$. The radius of curvature:

$$\rho = \frac{v^2}{a_n} = \frac{25}{4} = 6.25 \text{ m.}$$

Note: the physical radius of the helix in the xy -plane is 4 m, but the radius of curvature is larger (6.25 m) because the path also climbs in z , making it less sharply curved.

Meaning: "What We Just Did"

Acceleration is the change-in-a-blink of the **velocity vector**, not just the speed. A vector can change in two independent ways: magnitude (tangential) and direction (normal).

With algebra alone, $a_c = v^2/r$ is an approximation from similar triangles. The limit makes it exact. This decomposition of acceleration into tangential and normal components is one of the most powerful tools in mechanics — it turns any curved path into a local circle.



Basic Drills (*solutions in solutions/04-solutions.md*)

1. $r = 3 \text{ m}$, $v = 6 \text{ m/s}$. Find the centripetal acceleration.
2. $r = 50 \text{ m}$, $v = 10 \text{ m/s}$. Centripetal acceleration in m/s^2 and in g 's ($g = 10$).
3. $v(t) = 4t$, $\rho = 8 \text{ m}$. Find a_t , a_n , and total a at $t = 1$.
4. $\vec{r}(t) = (5 \cos 2t, 5 \sin 2t)$. Find $|\vec{a}|$ and confirm it equals v^2/r .
5. A car at constant speed 15 m/s takes a turn. The centripetal acceleration is 4.5 m/s^2 . What is the radius?



Advanced Drills (*solutions in solutions/04-solutions.md*)

1. A rollercoaster car at the top of a hill (radius 20 m) has speed 10 m/s. Find the centripetal acceleration. If $g = 10$, does the car leave the track?
2. A particle moves on $y = \frac{1}{2}x^2$ with constant x -speed $v_x = 2$ m/s. Find the acceleration vector at $x = 1$.
3. For $\vec{r}(t) = (t, t^2, t^3)$, find the radius of curvature at $t = 0$.
4. **Trap:** An object moves at constant speed v in a circle of radius r . If the radius doubles but the speed stays the same, what happens to the centripetal acceleration?
5. Derive the formula $a_n = \frac{|\vec{v} \times \vec{a}|}{|\vec{v}|}$ for the normal component of acceleration. Verify it gives the same answer as Example 5.



Intuitional Drills (*solutions in solutions/04-solutions.md*)

1. A car turns left at constant speed. Which way does the driver's body lean, and why?
2. A satellite in a circular orbit has constant speed. Is work being done on it by gravity?
3. A rollercoaster enters a circular loop of radius 10 m. At the **top** of the loop, what minimum speed keeps the car on the track?
4. Two identical cars take the same curve. Car A goes at 20 m/s, Car B at 40 m/s. How many times larger is the centripetal force needed by Car B?
5. A ball on a string is swung in a vertical circle. At which point in the circle is the tension in the string **greatest** — top or bottom?



Reading Physical Symbols

Symbol	How to Read It	Physical Meaning
$\hat{t}$	"t-hat"	The unit vector tangent to the path — points in the direction of motion at each instant.

Symbol	How to Read It	Physical Meaning
$\hat{n}$	"n-hat"	The unit vector normal to the path — points toward the center of curvature (the "inside" of the turn).
ρ	"rho"	Radius of curvature. The radius of the circle that best hugs the curve at a given point. Small ρ = sharp turn; $\rho = \infty$ = straight line.
$\frac{dv}{dt}$	"dee-v dee-t"	Tangential acceleration. How fast the <i>speed</i> is changing. Can be positive (speeding up) or negative (slowing down).
$\frac{v^2}{\rho}$	"v-squared over rho"	Normal (centripetal) acceleration. How hard the path is curving. Always points toward the inside of the turn.
$\times$ in $\vec{v} \times \vec{a}$	"cross"	The cross product extracts the perpendicular component. \$

Key habit: Any curved path can be thought of as a sequence of tangent circles. At each instant, find $\hat{t}$ (where you're going) and $\hat{n}$ (where you're turning toward). Acceleration lives in these two directions.

Glossary